Data Collection for Isolated Sign Language Recognition

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Introduction

We present the Panoptic View ASL Dataset, a collection of signing videos to train isolated sign language recognition technology. The dataset was filmed in a studio setup that simulated a 3D perspective with participants who self-identified as native or fluent signers and who use ASL in their daily lives. They were prompted to sign with a text display of individual letters, numbers, and basic daily signs. With a Deaf-led focus, data collection and analysis emphasized the importance of natural signing and ASL parameters.

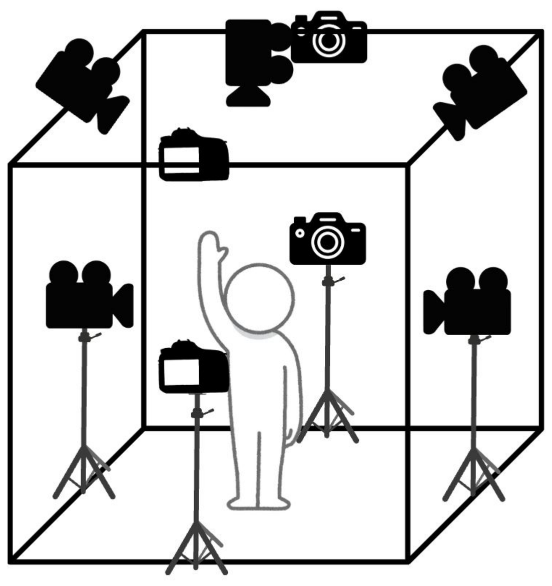
- Sign language recognition is a growing topic in across several disciplines
  - 988 publications on sign language recognition technology from the SCOPUS database between 2003 and 2023 [7]
- Popular machine learning approaches are CNNs and transformers with RGB or skeleton based input [3, 5, 6, 8]
- Large sign language datasets are developed using native signers and real-world recording environments [2, 4]
  - Data lacks robustness from various angles
  - Synthetic view augmentation success with cross-view recognition [5]
- Models neglect linguistically-motivated approaches [1]
- Research missing Deaf/fluent signer involvement [1]

Methods

MATERIALS

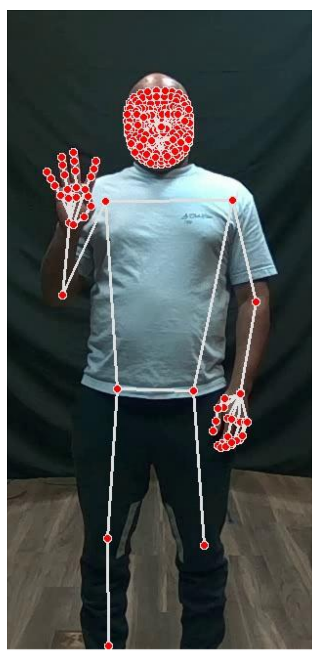
- Intel RealSense D455 Camera
  - depth, RGB, infrared sensors
  - 162° view

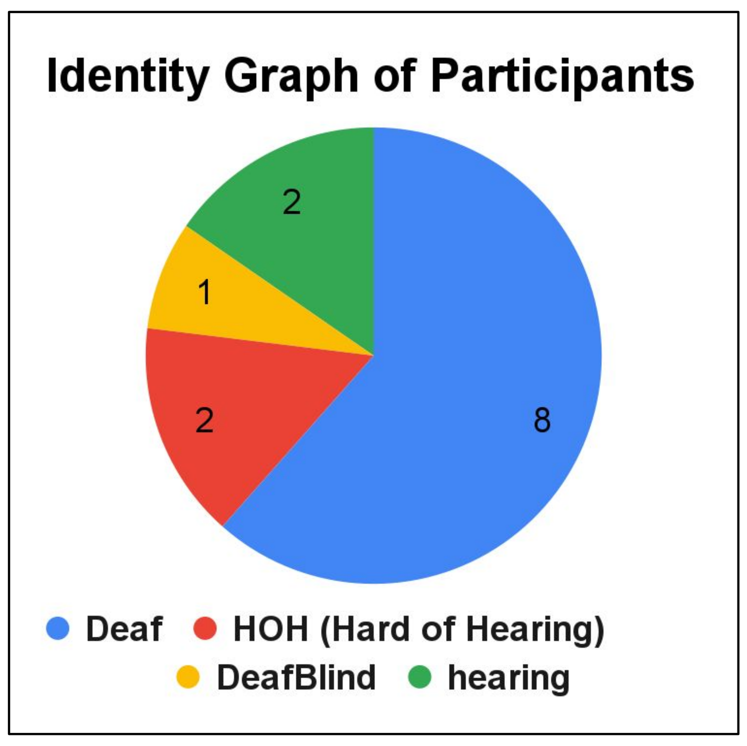
- Panoptic studio
  - 11x11 feet setup
  - 9 cameras: 4 at height 3ft, 4 at height 10 ft, 1 on ceiling
  - black background
- Code
  - Run cameras with PyRealSense and threading
  - Cut video using MediaPipe holistic landmarks



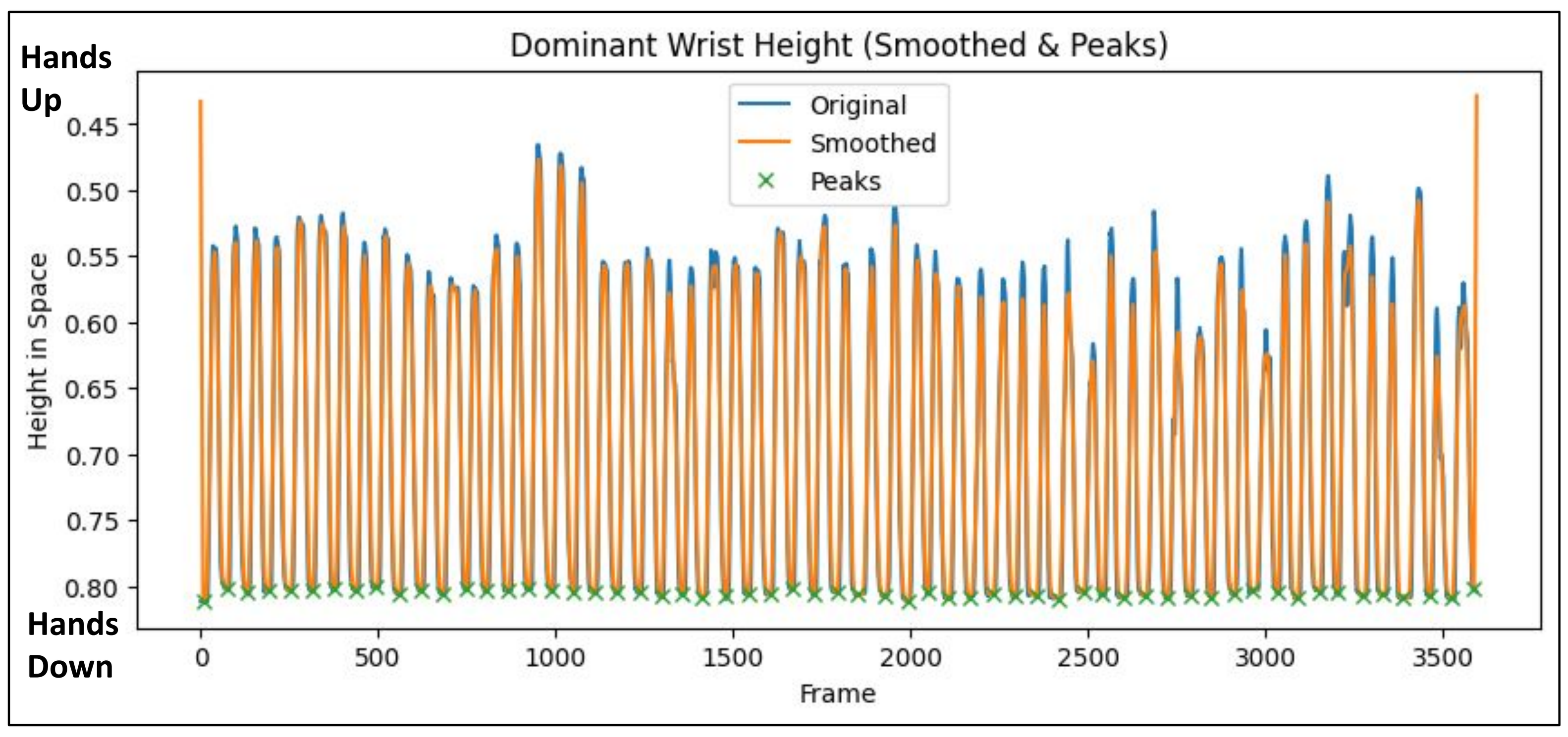
PROCEDURE

- Data collection
  - Preparation
    - Demographics survey
  - Prompting
    - 222 total letters, numbers, and basic signs
    - Text displayed for 2-4 seconds
    - Participants lower hand for each prompt
  - Recorded sitting and standing (if able)
- Data processing
  - Cut out non-signing
  - Splitting video by individual sign
    - Automatic
      - MediaPipe tracks wrist movements on front facing camera
      - Script to cut all camera angles by lowest hand placement
    - Semi-automatic
      - Manually verify automated cuts
    - Manual
      - Use ELAN to log timestamps and cut with script
  - Run MediaPipe to collect keypoints on individual camera and prompt videos





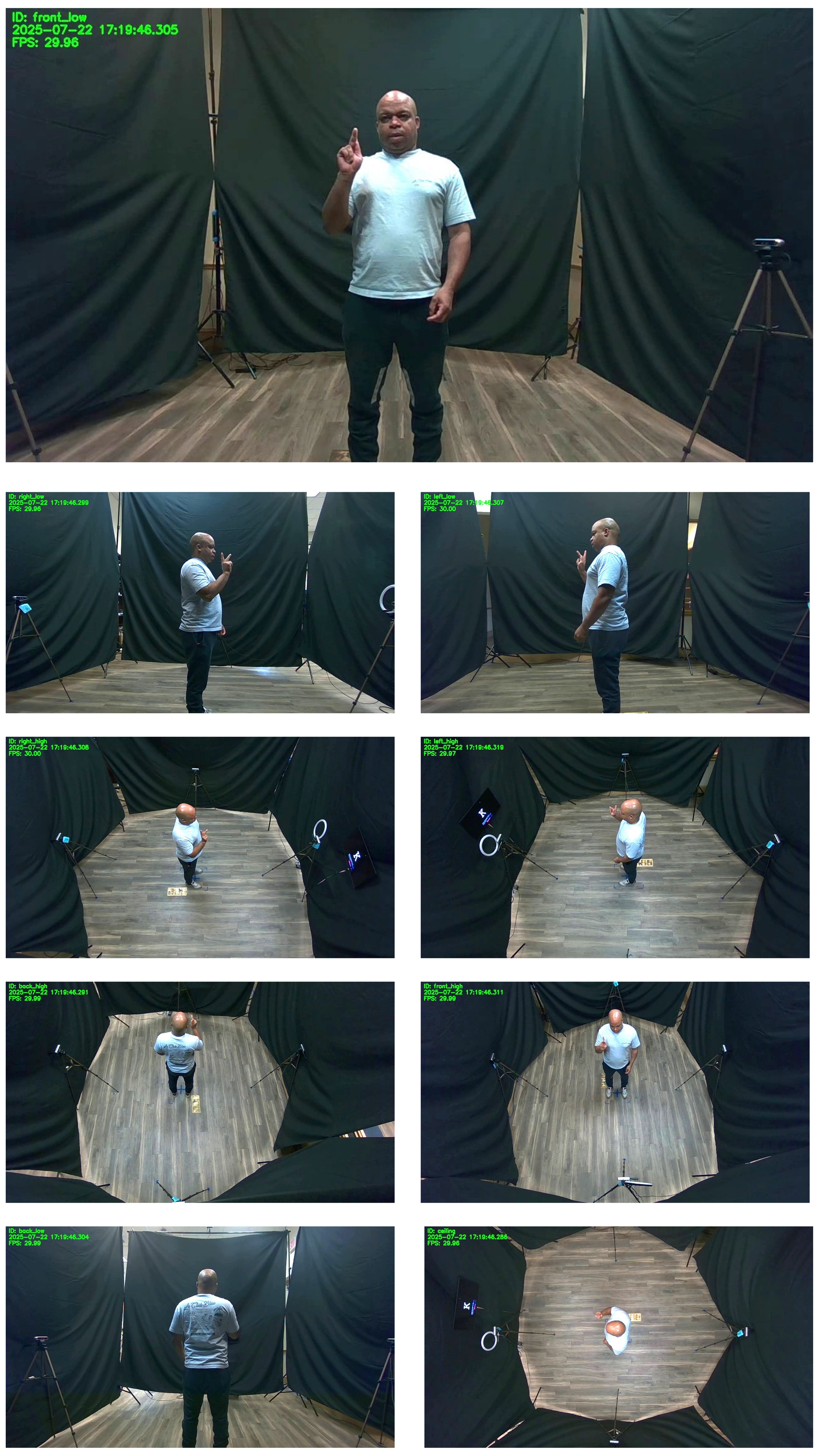
Self-reported hearing status of all participants



Height of dominant wrist across one recording for a participant (as tracked by MediaPipe). Changes in wrist height are associated with boundaries between individual sign responses and peaks are used for video segmentation.

Results

- Recorded 13 participants
  - Self-identified as native/fluent signers
- Lighting fixed for consistency
- Synchronized cameras with near perfect frame-level length
- Cut videos by sign using semi-automatic script
- Tested baseline transformer model on panoptic subset of A-Z and 4 fingerspelled signs
- Top-1 accuracy is as follows:
  - Front-Low: 60.61%
  - Front-High: 33.33%
  - Right-Low: 40.40%
  - Right-High: 16.16%
  - Left-Low: 48.48%
  - Left-High: 37.37%
  - Back-Low: 4.04%
  - Back-High: 6.06%
  - Ceiling: 6.06%
- Top-1 combined accuracy is as follows:
  - All angles: 38.50%
  - All but ceiling: 52.86%
  - Low (Front, Right, Left): 61.28%



Limitations

- Technology: computer and camera connection
- Studio Setup: lighting and background issues
- Human Prompting Errors: sign production variation between participants
- Logistics: unnatural sign prompting and production
- Time: recording, running code, and training model

Future work

This research paves the way in ASL panoptic view data and sets the expectation to follow community call to action in ISLR [1].

Apply data to isolated sign language recognition model and answer questions:

- How does applying an ASL-centered dataset and linguistically-informed modeling improve the results of panoptic view isolated sign language recognition models?
- How much does the quality and quantity of datasets affect the ability to produce an accurate ISLR model?
- How can personalized ISLR model enhance individual accuracy?

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